

printed circuit boards (PCBs) or multi-chip modules.

2. *Capacity.* No longer limited by the die size and defect density that Moore's Law dictates, multiple smaller dies can be inter-connected, and greater capacity and higher yield achieved than with a monolithic IC.

3. *Crossovers.* With 3D one can mix and match different types of dies (processors, memories, field-programmable gate arrays (FPGAs) and analogue) in order to produce system-on-chips (SoCs) that were never before possible.

Another similar packaging technology which is gaining steam is 2.5D, where die slices are mounted side-by-side on a passive silicon interposer substrate that is used just to connect the active dies. 2.5D offers many of the benefits of full 3D without challenges like complex design rules, co-design of the different stacked dies, heat dissipation and integration of through-silicon vias with deep submicron active devices.

Both 3D and 2.5D packaging technologies have their own advantages and disadvantages depending on the application. For instance, while 3D offers significant benefits, it also presents new design and manufacturing challenges. The precision alignment of through-silicon vias and solder microbumps is critical during assembly. The differing thermal coefficients of silicon substrates and microbumps cause reliability challenges because of mechanical stress and cracking.

Notwithstanding, both 3D and 2.5D are likely to have a long future.

Overall impact

Semiconductor manufacturing is very expensive. In general, for 300mm wafer sizes using leading-edge technology, foundries cost \$3-5 billion with capacities up to 40-50k per month. For memories, the cost is \$8-11 billion with capacities up to 200-250k per month. Equipment account for majority of the cost. Front-end equipment are much more expensive than back-end equipment as these add more value to the final semiconductor product.

'Fabless' semiconductor companies

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(which design chips but don't have their own manufacturing facilities) outsource the fabrication of the devices to a semiconductor foundry. The 'foundry' is a semiconductor fab that fabricates the designs of other companies.

Increasingly, there's an industry shift towards the foundry-based model. Instead of designing and manufacturing chips themselves, today, more companies choose to focus on only the chip design to avoid the expensive R&D and equipment costs associated with manufacturing. While this improves efficiencies, lowers costs and increases yields, it will also lead to even more industry consolidation. Currently, only fifteen companies manufacture devices at 32nm and below (this does not include R&D and pilot lines). This compares to 28 companies, which were 'leading edge' in 2001 (manufacturing on 200nm wafers—150nm and 130nm). And it is highly unlikely that any new company will enter the market.

The big issue for all the semiconductor manufacturing companies is "Where is the return on investment (ROI)?" The cost for 3D with through-silicon via (TSV) will be compared to the costs of the manufacturing facilities for the next node. Fab costs for the 22nm node are expected to approach \$6 billion, so few logic or memory integrated device manufacturers or foundries will be able to afford such costs.

By mid-2013 to early 2014, a preliminary 450mm R&D production line will be established at the College of Nanoscale Science and Engineering

of the University at Albany in New York. The facility will be the focus of the recently-announced Global 450 Consortium (G450C) efforts and is expected to contain 50 different tool types. The objective of the pilot line will be to develop data to support the purchase of production-line tools. Many believe that once 450mm enters full production, further 300mm node development will cease for tool suppliers due to limited resources and poor ROI.

Once 450mm is fully developed, spare capacity in 300mm will emerge, encouraging a migration from 200mm production, impacting the viability and competitiveness of both 200mm and 300mm fabs. In 15-20 years, even low-volume, mature technologies in micro-electromechanical systems (MEMS), power and analogue could migrate to 450mm fabs.

Advancements in 2.5D and 3D technologies will open up new market opportunities and enable a new level of system-level integration never before possible on silicon platforms. But there are still some hurdles to overcome for 3D IC design and manufacturing to become mainstream—both technical and business related. For example, the semiconductor industry must come together to set standards for design enablement, manufacturing, test and chip-to-chip interfaces.

The interposer-based active-on-passive 2.5D stacked silicon interconnect (SSI) technology could be a lower-risk alternative to full 3D—especially for FPGAs—that will offer power, cost and reliability advantages for many years to come.

Heading into 2012

The industry is at an exciting crossroads. All these developments are good for consumers, who thrive on 'smaller, faster, cheaper' chips which power the smaller, faster portable electronic devices that power the world we live in. The semiconductor industry will continue to deliver what the consumer wants and needs! ●

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